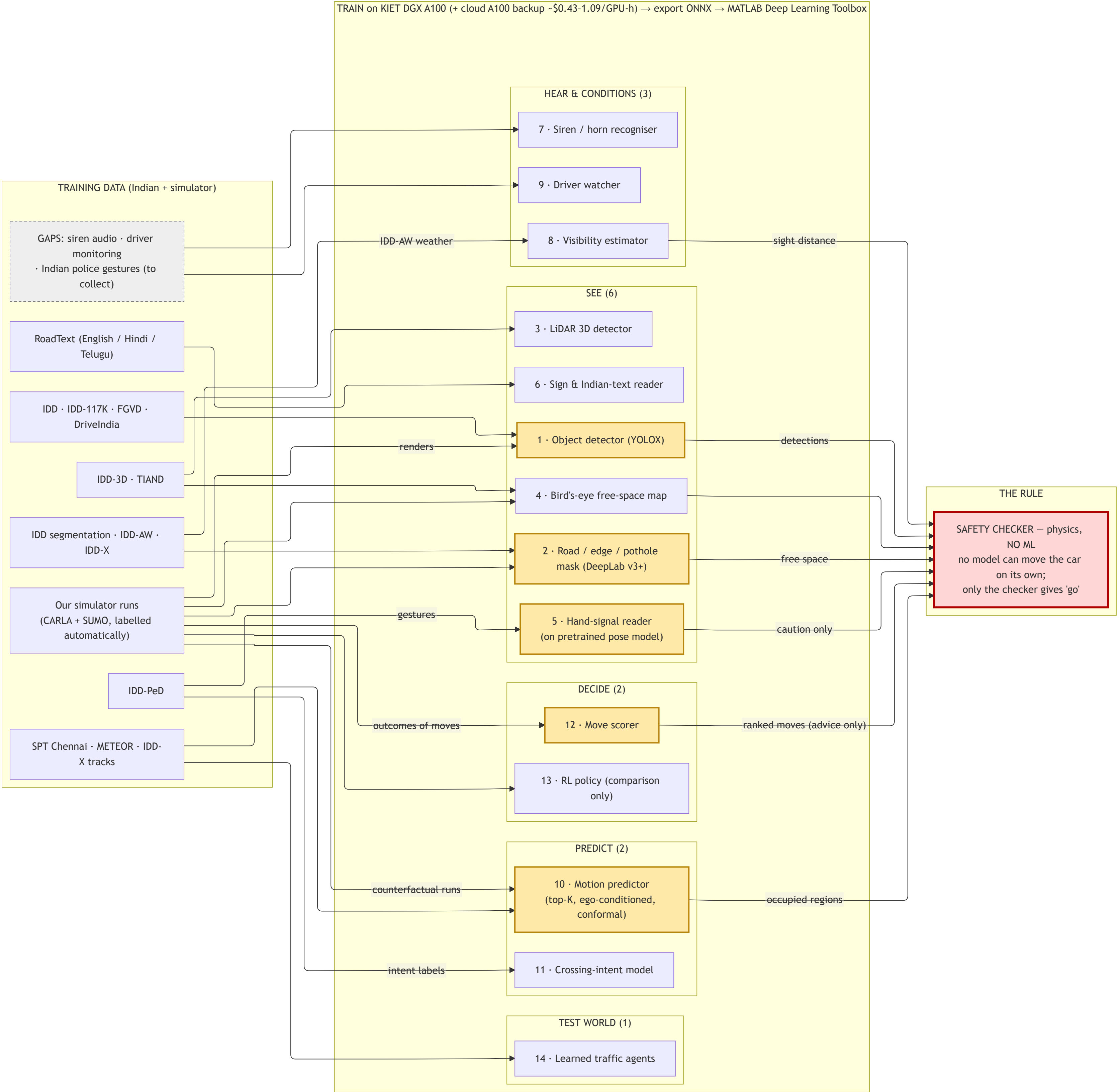


D3 · The 14-model AI stack (two tiers)

Gold = 5-model prototype core. White = full system (+9). Every model is advice only — the physics safety checker gives the only 'go'.



D3 · The 14-model AI stack (two tiers) — exact spec for redrawing

Redraw rule for Gemini / Nano Banana: keep every box, group and arrow listed. Do not add, merge or drop any. Styling is free; structure is fixed.

BOXES (grouped exactly as drawn):

- TRAINING DATA (Indian + simulator)
 - [d_idd] IDD · IDD-117K · FGVD · DriveIndia
 - [d_seg] IDD segmentation · IDD-AW · IDD-X
 - [d_3d] IDD-3D · TIAND
 - [d_ped] IDD-PeD
 - [d_traj] SPT Chennai · METEOR · IDD-X tracks
 - [d_text] RoadText (English / Hindi / Telugu)
 - [d_gap] GAPS: siren audio · driver monitoring · Indian police gestures (to collect)
 - [d_sim] Our simulator runs (CARLA + SUMO, labelled automatically)
- TRAIN on KIET DGX A100 (+ cloud A100 backup ~\$0.43-1.09/GPU-h) → export ONNX → MATLAB Deep Learning Toolbox
 - SEE (6)
 - [m1] 1 · Object detector (YOLOX)
 - [m2] 2 · Road / edge / pothole mask (DeepLab v3+)
 - [m3] 3 · LiDAR 3D detector
 - [m4] 4 · Bird's-eye free-space map
 - [m5] 5 · Hand-signal reader (on pretrained pose model)
 - [m6] 6 · Sign & Indian-text reader
 - HEAR & CONDITIONS (3)
 - [m7] 7 · Siren / horn recogniser
 - [m8] 8 · Visibility estimator
 - [m9] 9 · Driver watcher
 - PREDICT (2)
 - [m10] 10 · Motion predictor (top-K, ego-conditioned, conformal)
 - [m11] 11 · Crossing-intent model
 - DECIDE (2)
 - [m12] 12 · Move scorer
 - [m13] 13 · RL policy (comparison only)
 - TEST WORLD (1)
 - [m14] 14 · Learned traffic agents
- THE RULE
 - [gate] SAFETY CHECKER – physics, NO ML no model can move the car on its own; only the checker gives 'go'

CONNECTIONS (25 arrows, from → to : what flows):

IDD · IDD-117K · FGVD · DriveIndia → 1 · Object detector (YOLOX)
Our simulator runs (CARLA + SUMO, labelled automatically) → 1 · Object detector (YOLOX) : renders
IDD segmentation · IDD-AW · IDD-X → 2 · Road / edge / pothole mask (DeepLab v3+)
Our simulator runs (CARLA + SUMO, labelled automatically) → 2 · Road / edge / pothole mask (DeepLab v3+)
IDD-3D · TIAND → 3 · LiDAR 3D detector
IDD-3D · TIAND → 4 · Bird's-eye free-space map
Our simulator runs (CARLA + SUMO, labelled automatically) → 4 · Bird's-eye free-space map
IDD-PeD → 5 · Hand-signal reader (on pretrained pose model) : gestures
RoadText (English / Hindi / Telugu) → 6 · Sign & Indian-text reader
GAPS: siren audio · driver monitoring · Indian police gestures (to collect) → 7 · Siren / horn recogniser
IDD segmentation · IDD-AW · IDD-X → 8 · Visibility estimator : IDD-AW weather
GAPS: siren audio · driver monitoring · Indian police gestures (to collect) → 9 · Driver watcher
SPT Chennai · METEOR · IDD-X tracks → 10 · Motion predictor (top-K, ego-conditioned, conformal)
Our simulator runs (CARLA + SUMO, labelled automatically) → 10 · Motion predictor (top-K, ego-conditioned, conformal) : counterfactual runs
IDD-PeD → 11 · Crossing-intent model : intent labels
Our simulator runs (CARLA + SUMO, labelled automatically) → 12 · Move scorer : outcomes of moves
Our simulator runs (CARLA + SUMO, labelled automatically) → 13 · RL policy (comparison only)
SPT Chennai · METEOR · IDD-X tracks → 14 · Learned traffic agents
1 · Object detector (YOLOX) → SAFETY CHECKER – physics, NO ML no model can move the car on its own; only the checker gives 'go' : detections
2 · Road / edge / pothole mask (DeepLab v3+) → SAFETY CHECKER – physics, NO ML no model can move the car on its own; only the checker gives 'go' : free space
4 · Bird's-eye free-space map → SAFETY CHECKER – physics, NO ML no model can move the car on its own; only the checker gives 'go'
10 · Motion predictor (top-K, ego-conditioned, conformal) → SAFETY CHECKER – physics, NO ML no model can move the car on its own; only the checker gives 'go' : occupied regions
12 · Move scorer → SAFETY CHECKER – physics, NO ML no model can move the car on its own; only the checker gives 'go' : ranked moves (advice only)
5 · Hand-signal reader (on pretrained pose model) → SAFETY CHECKER – physics, NO ML no model can move the car on its own; only the checker gives 'go' : caution only
8 · Visibility estimator → SAFETY CHECKER – physics, NO ML no model can move the car on its own; only the checker gives 'go' : sight distance

Mermaid source (paste into mermaid.live to re-render)

```
graph LR
    subgraph DATA ["TRAINING DATA (Indian + simulator)"]
        d_idd["IDD · IDD-117K · FGVD · DriveIndia"]
        d_seg["IDD segmentation · IDD-AW · IDD-X"]
        d_3d["IDD-3D · TIAND"]
        d_ped["IDD-PeD"]
        d_traj["SPT Chennai · METEOR · IDD-X tracks"]
        d_text["RoadText (English / Hindi / Telugu)"]
        d_gap["GAPS: siren audio · driver monitoring<br/>· Indian police gestures (to collect)"]
        d_sim["Our simulator runs<br/>(CARLA + SUMO, labelled automatically)"]
    end
    subgraph TRAIN ["TRAIN on KIET DGX A100 (+ cloud A100 backup ~$0.43-1.09/GPU-h) → export ONNX → MATLAB Deep Learning Toolbox"]
        subgraph SEE ["SEE (6)"]
            m1["1 · Object detector (YOLOX)"]
            m2["2 · Road / edge / pothole mask (DeepLab v3+)"]
            m3["3 · LiDAR 3D detector"]
            m4["4 · Bird's-eye free-space map"]
            m5["5 · Hand-signal reader<br/>(on pretrained pose model)"]
            m6["6 · Sign & Indian-text reader"]
        end
        subgraph HEAR ["HEAR & CONDITIONS (3)"]
            m7["7 · Siren / horn recogniser"]
            m8["8 · Visibility estimator"]
            m9["9 · Driver watcher"]
        end
        subgraph PRED ["PREDICT (2)"]
            m10["10 · Motion predictor<br/>(top-K, ego-conditioned, conformal)"]
            m11["11 · Crossing-intent model"]
        end
        subgraph DEC ["DECIDE (2)"]
            m12["12 · Move scorer"]
            m13["13 · RL policy (comparison only)"]
        end
        subgraph WORLD ["TEST WORLD (1)"]
            m14["14 · Learned traffic agents"]
        end
    end
    subgraph GATE ["THE RULE"]
        gate["SAFETY CHECKER – physics, NO ML<br/>no model can move the car on its own;<br/>only the checker gives 'go'"]
    end
    d_idd --> m1
    d_sim --> renders["renders"]
    renders --> m1
    d_seg --> m2
    d_sim --> m2
    d_3d --> m3
    d_3d --> m4
    d_sim --> m4
    d_ped --> gestures["gestures"]
    gestures --> m5
    d_text --> m6
    d_gap --> m7
    d_seg --> IDD_AW_weather["IDD-AW weather"]
    IDD_AW_weather --> m8
    d_gap --> m9
    d_traj --> m10
    d_sim --> counterfactual_runs["counterfactual runs"]
    counterfactual_runs --> m10
    d_ped --> intent_labels["intent labels"]
    intent_labels --> m11
    d_sim --> outcomes_of_moves["outcomes of moves"]
    outcomes_of_moves --> m12
    d_sim --> m13
    d_traj --> m14
    m1 --> detections["detections"]
    detections --> gate
    m2 --> free_space["free space"]
    free_space --> gate
    m4 --> gate
    m10 --> occupied_regions["occupied regions"]
    occupied_regions --> gate
    m12 --> ranked_moves["ranked moves (advice only)"]
    ranked_moves --> gate
    m5 --> caution_only["caution only"]
    caution_only --> gate
    m8 --> sight_distance["sight distance"]
    sight_distance --> gate
    classDef core fill:#ffe9a8,stroke:#b8860b,stroke-width:2px
    class m1,m2,m5,m10,m12 core
    classDef safety fill:#ffd6d6,stroke:#b00000,stroke-width:3px
    class gate safety
    classDef gap fill:#eeeeee,stroke:#777,stroke-dasharray: 4 3
    class d_gap gap
```